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Q1. Food Delivery Billing System (Multiple Inheritance using Two Interfaces)

Score: 10.00 TC: 3/3 passed

CODE

```
import java.util.Scanner;

interface DeliveryCharge{
    double calculateDeliveryCharge(double foodCost);
}

interface CouponDiscount{
    double calculateDiscount(double foodCost, String couponCode);
}

class FoodOrder implements DeliveryCharge, CouponDiscount{

    public double calculateDeliveryCharge(double foodCost){
        if(foodCost >= 500)
            return 0;
        else
            return 50;
    }
    public double calculateDiscount(double foodCost, String couponCode){
        if (couponCode.equals("SAVE10"))
            return foodCost * 0.10;
        else
            return 0;
    }
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);

        double foodCost = sc.nextDouble();
        sc.nextLine();
        String couponCode = sc.nextLine();

        FoodOrder f = new FoodOrder();

        double deliveryCharge = f.calculateDeliveryCharge(foodCost);
        double discount = f.calculateDiscount(foodCost, couponCode);
        double finalBill = foodCost + deliveryCharge - discount;

        System.out.println("Food Cost : " + foodCost);
        System.out.println("Delivery Charge : " + deliveryCharge);
        System.out.println("Discount : " + discount);
        System.out.println("Final Bill : " + finalBill);

        sc.close();
    }
}
```

INPUT

600
SAVE10

OUTPUT

Food Cost : 600.0
Delivery Charge : 0.0
Discount : 60.0
Final Bill : 540.0

Q2. Hotel Billing System (Multiple Inheritance using One Class and One Interface)

Score: 10.00 TC: 3/3 passed

CODE

```
import java.util.Scanner;

interface ServiceCharge {
    double calculateServiceCharge();
}

class Room {
    int roomNumber;
    String guestName;
    double roomRent;

    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);

        HotelBill h = new HotelBill();

        h.roomNumber = sc.nextInt();
        sc.nextLine();
        h.guestName = sc.nextLine();
        h.roomRent = sc.nextDouble();

        double serviceCharge = h.calculateServiceCharge();
        double finalBill = h.roomRent + serviceCharge;

        System.out.println("Room Number : " + h.roomNumber);
        System.out.println("Guest Name : " + h.guestName);
        System.out.println("Room Rent : " + h.roomRent);
        System.out.println("Service Charge : " + serviceCharge);
        System.out.println("Final Bill : " + finalBill);
    }
}

class HotelBill extends Room implements ServiceCharge{

    public double calculateServiceCharge(){
        if (roomRent >= 5000)
            return roomRent * 0.12;
        else
            return roomRent * 0.08;
    }
}
```

INPUT

205
Ananya
6000

OUTPUT

Room Number : 205
Guest Name : Ananya
Room Rent : 6000.0
Service Charge : 720.0
Final Bill : 6720.0